Carl Boettiger

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Employment

2022-Present Associate Professor, Department of Environmental Science, Policy and Management, **University of California, Berkeley.**

2015-2022 Assistant Professor, Department of Environmental Science, Policy and Management, **University of California, Berkeley.**

2013-2015 NSF Post-doctoral Scholar in the Department of Applied Mathematics and Statistics, **University of California, Santa Cruz**. Mentors: Marc Mangel, Stephan Munch

Education

2012 Ph.D Population Biology, **University of California, Davis**. Mentor: Alan Hastings

2007 A.B in Physics, **Princeton University**, with honors and certificates in *biophysics* and *applied and computational mathematics.*

Publications

74. R. Quinn Thomas and **Carl Boettiger** (2026). Cyberinfrastructure to support ecological forecasting challenges. *Ecosphere* 17(6), e70682. doi:10.1002/ecs2.70682

73. Tasha Snow, Christopher Holdgraf, Wilson Sauthoff, Jessica Scheick, Ellianna Abrahams, Joanna Millstein, Sanjay Bhangar, **Carl Boettiger**, James Colliander, Luis A. Lopez, Elizabeth Holmes, Joseph H. Kennedy, Julia S. Lowndes, Alex I. Mandel, P. Yuvi, Fernando Pérez, Joe-Paul Swinski, Andy Teucher, and Matthew R. Siegfried (2026). A path to better science through co-creation and open infrastructure. *Perspectives of Earth and Space Scientists* 7(1), e2025CN000295. doi:10.1029/2025CN000295

72. Arnab Nandi, Wei-Lun Chao, Rongjun Qin, **Carl Boettiger**, Hilmar Lapp, and Tanya Berger-Wolf (2025). OmniMesh: Addressing findability challenges in distributed nature data repositories. *Proceedings of the 37th International Conference on Scalable Scientific Data Management (SSDBM)*, 1-6. doi:10.1145/3733723.3733728

71. Kari E. A. Norman, **Carl Boettiger**, Timothée Poisot, and Gavin M. Jones (2026). The role of AI in ecology's computational carbon footprint. *Frontiers in Ecology and the Environment*. doi:10.1002/fee.70021

70. Freya Olsson, Cayelan C. Carey, **Carl Boettiger**, and others (2025). What can we learn from 100,000 freshwater forecasts? A synthesis from the NEON Ecological Forecasting Challenge. *Ecological Applications*. doi:10.1002/eap.70004

69. Melissa Chapman, Martin Jung, David Leclère, **Carl Boettiger**, Andrey L. D. Augustynczik, Mykola Gusti, Leopold Ringwald, and Piero Visconti (2025). Meeting European Union biodiversity targets under future land-use demands. *Nature Ecology & Evolution*. doi:10.1038/s41559-025-02671-1

68. Robert L. Baker, Colin Smith, Sarah E. Wright, Issac Quevedo, Kristin Vanderbilt, **Carl Boettiger**, Judd M. Patterson, and Joe DeVivo (2025). NPSdataverse: a suite of R packages for data processing,

authoring Ecological Metadata Language metadata, checking data-metadata congruence, and accessing data. *The Journal of Open Source Software*. doi:10.21105/joss.08066

67. Caleb Scoville, Razvan Amironesei, Lily Xu, Melissa Chapman, Nick Record, and **Carl Boettiger** (2025). From maps to models: Participation and contestability in the dynamic management of natural resources. *Geo: Geography and Environment*. doi:10.1002/geo2.70028

66. Felipe Montealegre-Mora, **Carl Boettiger**, Carl J. Walters, and Christopher Cahill (2025). Using machine learning to inform harvest control rule design in complex fishery settings. *Fish and Fisheries*. doi:10.1111/faf.70013

65. Kari E. A. Norman, Perry de Valpine, and **Carl Boettiger** (2025). No General Trend in Functional Diversity in Bird and Mammal Communities Despite Compositional Change. *Global Ecology and Biogeography*. doi:10.1111/geb.13950

64. Abigail G. Keller, Timothy D. Counihan, Edwin D. Grosholz, **Carl Boettiger** (2025). The transition from resistance to acceptance: Managing a marine invasive species in a changing world. *Journal of Applied Ecology*. doi:10.1111/1365-2664.14881

63. Dietze, White, Abeyta, et al. (2024). Near-term ecological forecasting for climate change action. *Nature Climate Change* 14, 1236–1244. doi:10.1038/s41558-024-02182-0

62. Olsson, **Boettiger**, Carey, Lofton, Thomas (2024). Can you predict the future? A tutorial for the National Ecological Observatory Network Ecological Forecasting Challenge. *Journal of Open Source Education* 7(82), 259. doi:10.21105/jose.00259

61. Cattoni, South, Warne, **Boettiger**, Thakran, Holden (2024). Revisiting Fishery Sustainability Targets. *Bulletin of Mathematical Biology* 86, 127. doi:10.1007/s11538-024-01352-7

60. Pottinger, Geyer, Biyani, Martinez, Nathan, Morse, Liu, Hu, de Bruyn, **Boettiger**, Baker, McCauley (2024). Pathways to reduce global plastic waste mismanagement and greenhouse gas emissions by 2050. *Science* 386, 1168-1173. doi:10.1126/science.adr3837

59. Wenk, Sauquet, Gallagher, et al. (2024). The AusTraits plant dictionary. *Scientific Data* 11, 537. doi:10.1038/s41597-024-03368-z

58. Chapman, Goldstein, Schell, Brashares, Carter, Ellis-Soto, Faxon, Goldstein, Halpern, Longdon, Norman, O'Rourke, Scoville, Xu, **Boettiger** (2024). Biodiversity monitoring for a just planetary future. *Science* 383, 34-36. doi:10.1126/science.adh8874

57. Scoville, Faxon, Chapman, Fried, Xu, **Boettiger**, Reed, Lapeyrolerie, Van Scoyoc, Amironesei (2023). Environment, Society, and Machine Learning. In *The Oxford Handbook of the Sociology of Machine Learning*, C. Borch & JP. Pardo-Guerra (eds.), 527-548. Oxford University Press. doi:10.1093/oxfordhb/9780197653609.013.8

56. Dietze, Thomas, Peters, **Boettiger**, Koren, Shiklomanov, Ashander (2023). A community convention for ecological forecasting: Output files and metadata version 1.0. *Ecosphere* 14(11). doi:10.1002/ecs2.4686

55. Montealegre-Mora, Lapeyrolerie, Chapman, Keller, **Boettiger** (2023). Pretty Darn Good Control: When are Approximate Solutions Better than Approximate Models. *Bulletin of Mathematical Biology*. doi:10.1007/s11538-023-01198-5. (preprint, code)

54. David Moreau, Kristina Wiebels, and **Carl Boettiger** (2023). Containers for computational reproducibility. *Nature Reviews Methods Primers* 3 (50). doi:10.1038/s43586-023-00236-9

53. Melissa Chapman, Lily Xu, Marcus Lapeyrolerie, and **Carl Boettiger** (2023). Bridging adaptive management and reinforcement learning for more robust decisions. *Philosophical Transactions of the Royal Society B* 378, 2022-0195. doi:10.1098/rstb.2022.0195. (preprint)

52. Thomas, McClure, Moore, Woelmer, **Boettiger**, Figueiredo, Hensley, Carey (2023). Near-term forecasts of NEON lakes reveal gradients of environmental predictability across the U.S. *Frontiers in Ecology and the Environment* 21 (5) 220-226. doi:10.1002/fee.2623. (preprint, code, data)

51. Sydne Record, **Carl Boettiger**, and Christine R. Rollinson (2023). Synthesizing forecasts to inform decision-making and advance ecological theory. *Methods in Ecology and Evolution*, 14: 728-731. doi:10.1111/2041-210X.14070

50. Thomas, **Boettiger**, Carey, Dietze, Johnson, Kenney, McLachlan, Peters, Sokol, Weltzin, Willson, Woelmer, and Challenge Contributors (2023). The NEON Ecological Forecasting Challenge. *Frontiers in Ecology and the Environment* 21 (3) 112-113. doi:10.1002/fee.2616. (preprint)

49. Chapman, **Boettiger**, Brashares (2023). Leveraging private land conservation to meet 2030 biodiversity targets in the United States. *Conservation Science and Practice*. doi:10.1111/csp2.12897. (preprint, code)

48. Benjamin S. Halpern, **Carl Boettiger**, Michael C. Dietze, Jessica A. Gephart, Patrick Gonzalez, Nancy B. Grimm, et al. (116 co-authors) (2023). Priorities for synthesis research in ecology and environmental science. *Ecosphere*. doi:10.1002/ecs2.4342

47. Lapeyrolerie, **Boettiger** (2022). Limits to Ecological Forecasting: Estimating Uncertainty for Critical Transitions with Deep Learning. *Methods in Ecology and Evolution*. doi:10.1111/2041-210X.14013. (preprint, code)

46. Lapeyrolerie, Chapman, Norman, **Boettiger** (2022). Deep Reinforcement Learning for Conservation Decisions. *Methods in Ecology and Evolution*. doi:10.1111/2041-210X.13954. (preprint, code)

45. **Carl Boettiger** (2022). The Forecast Trap. *Ecology Letters* 25, 1655-1664. doi:10.1111/ele.14024. (preprint, code)

44. Chapman, Wiltshire, Baur, Bowles, Carlisle, Castillo, Esquivel, Gennet, Iles, Karp, Kremen, Liebert, Olimpi, Ory, Ryan, Sciligo, Thompson, Waterhouse, **Boettiger** (2022). Social-ecological feedbacks drive tipping points in farming system diversification. *One Earth*. doi:10.1016/j.oneear.2022.02.007. (preprint, code)

43. Chapman, Scoville, Lapeyrolerie, **Boettiger** (2021). Power and Accountability in Reinforcement Learning Applications to Environmental Policy. *35th Conference on Neural Information Processing Systems (NeurIPS 2021)*. (preprint, pdf)

42. Lapeyrolerie, **Boettiger** (2021). Teaching machines to anticipate catastrophes. *Proceedings of the National Academy of Sciences*. doi:10.1073/pnas.2115605118

41. Chapman, Oestreich, Frawley, **Boettiger**, Diver, Santos, Scoville, Armstrong, Blondin, Chand, Haulsee, Knight, Crowder (2021). Promoting equity in the use of algorithms for high seas conservation. *One Earth*. doi:10.1016/j.oneear.2021.05.011. (preprint, code)

40. Karatayev, Baskett, Kushner, Shears, Caselle, **Boettiger** (2021). Grazer behavior can regulate large-scale patterning of community states. *Ecology Letters*. doi:10.1111/ele.13828. (preprint)

39. Reimer, Arroyo-Esquivel, Jiang, Scharf, Wolkovich, Zhu, **Boettiger** (2021). Noise can create or erase long transient dynamics. *Theoretical Ecology*. doi:10.1007/s12080-021-00518-6. (code, data)

38. Caleb Scoville, Melissa Chapman, Razvan Amironesei, **Carl Boettiger** (2021). Algorithmic conservation in a changing climate. *Current Opinion in Environmental Sustainability* 51, 30-35. doi:10.1016/j.cosust.2021.01.009.

37. **Carl Boettiger** (2020). [Rp] Fluctuation domains in adaptive evolution. *ReScience C* 6, 1, #15, doi:10.5281/zenodo.4081202. (pdf, code)

36. Pascal, Memarzadeh, **Boettiger**, Lloyd, Chadès (2020). A Shiny r app to solve the problem of when to stop managing or surveying species under imperfect detection. *Methods in Ecology and Evolution*. doi:10.1111/2041-210X.13501 (software)

35. Kari Norman, Scott Chamberlain, **Carl Boettiger** (2020). taxadb: A high-performance local taxonomic database interface. *Methods in Ecology and Evolution*. doi:10.1111/2041-210X.13440 (software)

34. **Carl Boettiger** (2020). Ecological management of stochastic systems with long transients. *Theoretical Ecology*. doi:10.1007/s12080-020-00477-4. (code)

33. de Aguiar, Newman, Pires, Yeakel, **Boettiger**, Burkle, Gravel, Guimarães Jr, O'Donnell, Poisot, Fortin, Hembry (2019). Revealing biases in the sampling of ecological interaction networks, *PeerJ*. doi:10.7717/peerj.7566. (software).

32. Milad Memarzadeh, Gregory L. Britten, Boris Worm, **Carl Boettiger** (2019). Rebuilding global fisheries under uncertainty. *Proceedings of the National Academy of Sciences*. doi:10.1073/pnas.1902657116

31. **Carl Boettiger**, Ryan Batt (2019). Bifurcation or state tipping: assessing transition type in a model trophic cascade. *Journal of Mathematical Biology*. doi:10.1007/s00285-019-01358-z (preprint, code)

30. Milad Memarzadeh, **Carl Boettiger** (2019). Resolving the Measurement Uncertainty Paradox in Ecological Management. *American Naturalist*. doi:10.1086/702704. (preprint, code, data). F1000 recommended

29. Dan Sholler, Karthik Ram, **Carl Boettiger**, Daniel S. Katz (2019). Enforcing public data archiving policies in academic publishing: A study of ecology journals. *Big Data & Society* 6(1) 1-18. doi:10.1177/2053951719836258. (preprint)

28. **Carl Boettiger** (2019). Ecological Metadata as Linked Data. *Journal of Open Source Software*, 4(34), 1276, doi:10.21105/joss.01276 (software).

27. Katz, Allen, Barba, Berg, Bik, **Boettiger**, et al. (32 co-authors) (2018). The principles of tomorrow's university. *F1000Research*, 7:1926. doi:10.12688/f1000research.17425.1.

26. Karthik Ram, **Carl Boettiger**, Scott Chamberlain, Noam Ross, Maëlle Salmon, and Stephanie Butland (2018). A Community of Practice Around Peer-review for Long-term Research Software Sustainability. *Computing in Science & Engineering*, 9615(c), 1–1. doi:10.1109/MCSE.2018.2882753

25. **Carl Boettiger** (2018). Managing Larger Data on a GitHub Repository. *Journal of Open Source Software*, 3(29), 971, doi:10.21105/joss.00971. (software).

24. Milad Memarzadeh, **Carl Boettiger** (2018). Adaptive management of ecological systems under partial observability. *Biological Conservation* 224, 9-15. doi:10.1016/j.biocon.2018.05.009. (software)

23. **Carl Boettiger** (2018). From noise to knowledge: how randomness generates novel phenomena and reveals information. *Ecology Letters*. doi:10.1111/ele.13085 (preprint, code, data)

22. **Carl Boettiger**, Dirk Eddelbuettel (2018). An Introduction to Rocker: Docker Containers for R. *The R Journal*. doi:10.32614/RJ-2017-065

21. **Carl Boettiger** (2017). Generating Codemeta Metadata for R Packages. *The Journal of Open Source Software* 2 (19), 454, doi:10.21105/joss.00454

20. Getz, Marshall, Carlson, Giuggioli, Ryan, Romañach, **Boettiger**, Chamberlain, Larsen, D'Odorico, O'Sullivan (2017). Making ecological models adequate. *Ecology Letters*. doi:10.1111/ele.12893

19. Ben Marwick, **Carl Boettiger**, Lincoln Mullen (2017). Packaging data analytical work reproducibly using R (and friends). *The American Statistician*. doi:10.1080/00031305.2017.1375986. (preprint)

18. Hampton, Jones, Wasser, Schuldhauer, Supp, Brun, Hernandez, **Boettiger**, Collins, Gross, Fernandez, Budden, White, Teal, Labou, Aukema (2017). Skills and Knowledge for Data Intensive Research. *BioScience*. doi:10.1093/biosci/bix025. (preprint)

17. T. Alex Perkins, **Carl Boettiger**, Benjamin L. Philips (2016). After the games are over: life-history trade-offs drive dispersal attenuation following range expansion. *Ecology and Evolution* 6 (18) 6425-6434. doi:10.1002/ece3.2314. (preprint, code)

16. **Carl Boettiger**, Michael Bode, James N. Sanchirico, Jacob LaRiviere, Alan Hastings, and Paul Robert Armsworth (2016). Optimal management of a stochastically varying population when policy adjustment is costly. *Ecological Applications* 26 (3) 808-817. doi:10.1890/15-0236. (preprint, code)

15. **Carl Boettiger**, Scott Chamberlain, Rutger Vos and Hilmar Lapp (2016). RNeXML: a package for reading and writing richly annotated phylogenetic, character, and trait data in R. *Methods in Ecology and Evolution*. doi:10.1111/2041-210X.12469. (preprint, code, software)

14. **Carl Boettiger**, Scott Chamberlain, Edmund Hart, Karthik Ram (2015). Building Software, Building Community: Lessons from the rOpenSci Project. *Journal of Open Research Software* 3: e8, doi:10.5334/jors.bu.

13. **Carl Boettiger** (2015). An introduction to Docker for reproducible research. *ACM SIGOPS Operating Systems Review* 49(1), 71-79. doi:10.1145/2723872.2723882. (preprint)

12. **Carl Boettiger**, Marc Mangel, Stephan Munch (2015). Avoiding tipping points in fisheries management through Gaussian process dynamic programming. *Proceedings of the Royal Society B* 282(1801), 8–11. doi:10.1098/rspb.2014.1631. (preprint, code, data). F1000 recommended

11. **Carl Boettiger**, Alan Hastings (2013). No early warning signals for stochastic transitions: insights from large deviation theory. *Proceedings of the Royal Society B*. doi:10.1098/rspb.2013.1372. (preprint, code)

10. **Carl Boettiger**, Noam Ross, Alan Hastings (2013). Early warning signals: The charted and uncharted territories. *Theoretical Ecology*. doi:10.1007/s12080-013-0192-6. (preprint, code)

9. **Carl Boettiger**, Alan Hastings (2013). Tipping points: From patterns to predictions, *Nature* 493, 157–158. doi:10.1038/493157a.

8. **Carl Boettiger**, Alan Hastings (2012). Early Warning Signals and the Prosecutor's Fallacy, *Proceedings of the Royal Society B* 279 (1748) 4734-4739. doi:10.1098/rspb.2012.2085. (preprint, code, data)

7. **Carl Boettiger**, Duncan Temple Lang, Peter Wainwright (2012). rfishbase: exploring, manipulating and visualizing FishBase data from R, *Journal of Fish Biology* 81 (6) 2030–2039. doi:10.1111/j.1095-8649.2012.03464.x. (code, software)

6. **Carl Boettiger**, Duncan Temple Lang (2012). Treebase: An R package for discovery, access and manipulation of online phylogenies, *Methods in Ecology and Evolution* 3 (6) 1060–1066. doi:10.1111/j.2041-210X.2012.00247.x. (code, software)

5. **Carl Boettiger**, Alan Hastings (2012). Quantifying Limits to Detection of Early Warning for Critical Transitions, *Journal of the Royal Society: Interface* 9 (75) 2527-2539. doi:10.1098/rsif.2012.0125. (preprint, code)

4. Jeremy M. Beaulieu, Dwueng-Chwuan Jhwueng, **Carl Boettiger**, Brian O'Meara (2012). Modeling Stabilizing Selection: Expanding the Ornstein-Uhlenbeck Model of Adaptive Evolution, *Evolution* 66 (8) 2369-2383. doi:10.1111/j.1558-5646.2012.01619.x. (software)

3. **Carl Boettiger**, Graham Coop, Peter Ralph (2012). Is your phylogeny informative? Measuring the power of comparative methods, *Evolution* 66 (7) 2240-51. doi:10.1111/j.1558-5646.2011.01574.x. (preprint, code, data)

2. **Carl Boettiger**, Jonathan Dushoff, Joshua S. Weitz (2010). Fluctuation domains in adaptive evolution, *Theoretical Population Biology* 77 (1) 6-13. doi:10.1016/j.tpb.2009.10.003. (preprint, code, data)

1. James J. Wray, Neta A. Bahcall, Paul Bode, **Carl Boettiger**, Philip Hopkins (2006). The Shape, Multiplicity, and Evolution of Superclusters in Lambda-CDM Cosmology, *The Astrophysical Journal* 652 (2) 907-916. doi:10.1086/508600. (preprint)

Book Chapters

Carl Boettiger (2017). A Reproducible R Notebook Using Docker. In J. Kitzes, D. Turek, & F. Deniz (Eds.), The Practice of Reproducible Research: Case Studies and Lessons from the Data-Intensive Sciences (1st ed., pp. 109–117). Oakland, CA: UC Press. `https://www.ucpress.edu/book.php?isbn=9780520294752`

Grants

Leveraging NASA Data to Guide Biodiversity Conservation Investments with The Trust for Public Land. **NASA National Aeronautics and Space Administration** (2024-2027). Federal Award 80NSSC25K7241. Carl Boettiger, PI. Total: $615,116.

The Eric and Wendy Schmidt Center for Data Science and Environment. (2023-2028) Douglas McCauley, Fernando Perez, Justin Brashares, Carl Boettiger Total: $12.6M.

Classroom and research access to GPU computing on the National Research Platform. **NSF National Artificial Intelligence Research Resource (NAIRR) Pilot** (2024 - current). Carl Boettiger, PI. Allocation of computational resources.

Examining Environmental Justice through Open Source, Cloud Native Tools. **NASA National Aeronautics and Space Administration** (2022-2025). UCB Award ID: 055956-001. Carl Boettiger, PI. Total: $134,596.

Collaborative Research: Frameworks: DeCODER (Democratized Cyberinfrastructure for Open Discovery to Enable Research). (2022-2026) **National Science Foundation** #OAC-2209865, UCB sub-award ID: 054904-001; PI $102,984

A decision framework for managing European Green Crab infestations on the coast of Washington and Salish Sea shorelines. **US Geological Survey** (2022 - 2025). UCB Award ID: 054356-001. Carl Boettiger, PI. Total: $174,577.

CAREER: Harnessing the data revolution for predicting and managing ecosystem regime shifts. Carl Boettiger. (2020-2025). **National Science Foundation** #DBI-1942280. $504,335.00

The Rocker Project. Carl Boettiger, Noam Ross, Dirk Eddelbuettel. (2019 - 2021). **Chan-Zuckerberg Initiative**: Essential Open Source Software for Science. $75,912

The Influence of Conflicting Policies and Supply-Chain Pressures on Farmers' Decisions and Tradeoffs with Respect to Biodiversity, Profitability, and Sustainability. Timothy Bowles (PI), Alastair Iles, Claire Kremen, Carl Boettiger. (2018-2022). **National Science Foundation** #CNH-1824871 $1,301,737

The rOpenSci Project. Karthik Ram, Carl Boettiger, Scott Chamberlain. (renewal, 2019-2021). **Helmsley Charitable Trust**, award 2016PG-BRI004. $1,000,000

Detecting Change in Global Biodiversity through Large Scale Network Analysis. Carl Boettiger, Rosemary Gillespie, Rasmus Nielsen. (2018). Berkeley Institute for Data Science, $67,000

Big Data, Big Uncertainty: Ecological Decision- Making in the 21st Century. (2018-2020). Hellman Fellows Award, **The Hellman Foundation**. $37,600

Berkeley Collegium Award for Narrowing the Gap Between Teaching and Research. (2018-2019) $16,867.50

Managing ecosystems under extreme uncertainty. (2016 - 2019) NSF **XSEDE** TG-DEB160003. NSF Estimated value of computing resources: $34,558

Reproducible and Collaborative Data Science. NSF **XSEDE** TG-DEB160021. (2016 - 2019) NSF Estimated value of computing resources: $20,028

James S McDonnell Foundation Post-doctoral Fellowship in Complex Systems (Awarded to post-doctoral scholar Allison Barner, who then chose to bring this award to my group at UC Berkeley) $200,000.

The rOpenSci Project (2015-2018, Co-PI). Karthik Ram, Carl Boettiger, Scott Chamberlain. **Helmsley Charitable Trust**, award 2016PG-BRI004. $2,875,071

The CodeMeta Project (2015). **National Science Foundation** #ACI-1549758 *$165,782*. (proposal link)

The rOpenSci Project, Phase II funding (2014, Co-PI). Karthik Ram, Carl Boettiger, Scott Chamberlain. **Alfred P. Sloan Foundation** $300,000

NSF Biology Post-doc (2013-2015). **National Science Foundation** #DBI-1306697, $138,000 (proposal link)

The rOpenSci Project Phase I funding (2013 Co-PI). Karthik Ram, Carl Boettiger, Scott Chamberlain. **Alfred P. Sloan Foundation** $180,000

IIASA YSSP fellowship (2009). **National Academy of Sciences**, OISE-0738129, $8,000. (proposal link)

Computational Science Graduate Fellowship (2008). **United States Department of Energy**, #DE-FG02-97ER25308, $149,000. (proposal link)

Teaching

ESPM-157: Data Science for Global Change Ecology. UC Berkeley. Upper-division undergraduate course, 4 units. Offered every Fall.

ESPM-288: Reproducible and Collaborative Data Science. UC Berkeley. Graduate-level course, 3 units. Offered every Spring.

CSOL-208: Data Science for Climate Solutions. UC Berkeley. Graduate-level course in the Masters of Climate Solutions, 3 units. Created 2026; offered every Spring.

ESPM-88B: Data Science in Ecology and the Environment. UC Berkeley. Freshman level Data-8 connector; not currently offered.

Invited Talks & Workshops

2026

NASA Biodiversity and Ecological Conservation Teams meeting, Washington, DC.
Invited seminar speaker, Environmental Informatics Hub at Monash University, Melbourne, Australia.
Natural Capital Project Symposium, Stanford, CA.
Plenary speaker, California Interagency Ecological Program Annual Workshop, Sacramento, CA.
Beyond Yellowstone Partners Meeting, Laramie, WY.
Invited speaker, AI-Ready Ecology and Biodiversity Data Infrastructure, NCEAS, Santa Barbara, CA.
National Geographic Blue Boundaries Wetlands, San Francisco, CA.

2025

ESIIL NSF Synthesis Center Summit, invited speaker, Boulder, CO.

Invited speaker, California 30x30 Partnership Summit, San Diego, CA.
NASA Biodiversity and Ecological Conservation Teams meeting, Washington, DC.
Invited NSF workshop on AI, Sensors, and Ecological Infrastructure, Pittsburgh, PA.
Invited workshop, Sentinel Sites Biodiversity, UC Davis, Davis, CA.
Urban Biodiversity Alliance, San Diego Museum of Natural History, San Diego, CA.
Invited speaker, Serrapilheira Institute Quantitative Ecology, Brazil.
Invited seminar speaker, Ocean Sciences, UC Santa Cruz, Santa Cruz, CA.

2024

Invited speaker, National Artificial Intelligence Resource Research (NAIRR) Pilot conference, Argonne National Labs, Chicago, IL.
Invited speaker, Conservation Data Justice Conference, Barcelona, Spain.
Invited speaker, NSF Cyberinfrastructure for Ecological Forecasting Conference, Portsmouth, NH.
NASA TOPS symposium, Kennedy Space Center, FL.

2023

Bezos Earth Fund AI for Climate and Nature Workshop, San Francisco, CA (recording).
Invited seminar speaker, University of Southern California, Biology Department, CA.
Invited seminar speaker, University of California, Davis, Department of Mathematics, CA.
Invited seminar speaker, Virginia Polytechnic Institute and State University (Virginia Tech), VA.
Invited workshop speaker, Critical Transitions, Princeton University, NJ.
Invited seminar speaker, Data Sciences, University of New South Wales, Sydney, Australia.
Invited talk, Gordon Conference on Predictive Ecology, Stonehill College, MA.
Co-organizer, Ecological Forecasting Unconference, NEON Headquarters, Boulder, CO.
Invited workshop speaker, Australian Mathematical Sciences (AMSI) Winter School, Brisbane, Australia.
Invited seminar speaker, University of Queensland, Brisbane, Australia.
Ecological Society of America, contributed talk, Portland, OR.
GEO BON Global Conference: Monitoring Biodiversity for Action, Montreal, Canada.
California Academy of Natural Sciences Past, Present, Future Biodiversity Convening, San Francisco, CA.

2022

Ecological Forecasting Initiative workshop (online).
Invited speaker, International Institute of Applied Systems Analysis (IIASA) conference, Boston University, Boston, MA.

2021

Workshop on the Future of Synthesis, National Center for Ecological Analysis and Synthesis (co-organizer).
National Ecological Observatory Network (NEON) Workshop on Complex Landscapes at Scale.
Ecological Forecasting Initiative (EFI) Workshop on Empowering Development of the Next Generation of Educational Materials for Forecasting.

2020

Invited seminar speaker, University of Oregon, Eugene, OR.
Chan-Zuckerberg Initiative Essential Open Source Software for Science Workshop.
Invited seminar speaker, DataONE data repository network.
Ecological Forecasting Initiative Research Coordination Network Workshop.
Invited seminar speaker, Concordia University, Montreal, Canada.

2019

Invited Biodiversity Research Center seminar speaker, University of British Columbia, Vancouver, Canada.

Ecological Forecasting Oral Session, American Geophysical Union, San Francisco, CA.
Advancing Theory in Ecology, NSF Workshop. Pennsylvania State University, State College, PA.
Biodiversity Data Workshop, invited speaker, Arizona State University, Tempe, AZ.
Transients in Ecology, Organized Oral Session, Ecological Society of America Annual Meeting, Louisville, KY.
Ecological Forecasting Initiative Conference, AAAS Headquarters, Washington, DC.
NIMBioS Transient Dynamics Workshop, University of Tennessee, Knoxville, TN.
Project EDDIE keynote speaker, Carleton College, Northfield, MN.
US Research Software Sustainability Institute Workshop, NCEAS, Santa Barbara, CA.
COMPASS Workshop for scientific communication, Asilomar, CA.

2018

NSF SI2 PIs Meeting, Washington, DC.
rOpenSci unconference, Seattle, WA.
Faculty Learning Program Fellows Workshop Berkeley, CA.
GraphXD, Berkeley, CA.
Digital Data in Biodiversity UC Berkeley. (co-organizer).
Ecological Society of America Invited Symposium Addressing Outstanding Challenges to Operationalizing Resilience (Organizer). New Orleans, LA.
Nonlinear Forecasting for Fisheries Applications, NOAA Southwest Fisheries Science Center, Santa Cruz, CA.

2017

CROSS Symposium Speaker Santa Cruz, CA.
Imagining Tomorrow's University Chicago, IL.
rOpenSci unconference, Los Angeles, CA.
Prov-a-thon: Practical Tools for Reproducible Science, Tamaya, NM.
NSF Translational Data Science Workshop, Berkeley Institute for Data Science, Berkeley, CA.

2016

Force16 CodeMeta Workshop, Portland, OR (organizer).
CodeMeta NSF Workshop Portland, OR (organizer).
rOpenSci unconference, San Francisco, CA.

2015

Empirical Dynamical Modeling and Forecasting in Nonlinear Systems, NTU, Taiwan.
Moore-Sloan Data Science Environments: Second Annual Data Science Summit (Workshop).
Data Intensive Training Workshop, NCEAS, Santa Barbara, CA.
NSF Big Data Hubs Design Charette, Western Region.
rOpenSci unconference, San Francisco, CA.
Pretty Darn Good Control Working group, NIMBioS, Knoxville, TN.

2014

Berkeley Initiative for Global Change Biology Workshop (student organized), UC Berkeley, CA.
DIMACS Global Change, Berkeley, CA.
Zoology Seminar, University of Wisconsin, Madison, WI.
ESPM Seminar, UC Berkeley, CA.
rOpenSci unconference, San Francisco, CA.
Reproducible Science: Curriculum & Workflow Workshop, NESCent, Durham, NC.
WSSSPE 2.0 Meeting. New Orleans, LA.

Workflows Working Group, NCEAS, Santa Barbara, CA.

2013 & prior

invited speaker, MBI, Sustainable Management of Living Natural Resources, Columbus, OH.
invited seminar speaker, WHOI, Woods Hole, MA.
invited seminar speaker, UC Davis Dept of Environmental Resources and Economics; Davis, CA.
invited speaker, SSB Symposium, Evolution Conference, Ottawa, Canada.
Sustainable Management of Living Natural Resources Workshop, MBI Columbus, OH.
Academic software & workforce development Workshop, ISEES. Oakland, CA.
Software Lifecycle Workshop, ISEES, Santa Barbara, CA.
Pretty Darn Good Control Working group, NIMBioS, Knoxville, TN.
Stochastic spatial modeling in population dynamics Workshop, AIM, Palo Alto, CA.

Awards

2020 CAREER Award, National Science Foundation
2020 Early Career Fellow, Ecological Society of America
2020 Winkler Family Foundation Scholar
2018 Hellman Fellow, Hellman Foundation
2011 Volterra Award, Ecological Society of America, Theory Section
2007 Elected to Membership in the Society of Sigma Xi
2007 Allen G. Shenstone Prize in Physics, Princeton University
2007 The Class of 1870 Old English Prize, Princeton University
2006 Kusaka Memorial Prize in Physics, Princeton University

Service & Outreach

Campus

Aug 2025 - May 2026. ESPM Equity Advisor, and Co-chair, Justice, Equity, Diversity, Inclusion & Belonging (JEDI-B) Committee
Aug 2025 - May 2026. ESPM Council
Jan 2026 - May 2026. Admissions Chair, Organisms & Environment Division, ESPM
Jan 2026 - May 2026. ESPM Graduate Programs Council
Spring 2026 - current. Founder and administrator, ESPM department computational resource (GPU and large-memory Jupyter/RStudio service)
2025 - 2026. Internal Review Committee, Data Science Undergraduate Studies programs
Jan 2025 - May 2025. CDSS ad hoc Committee on the Interdisciplinary Data Sciences proposal
2023 - 2024. RCNR Public-Private Partnership Proposals
2023 - 2024. CDSS College Regulations Proposal Task Force
2023 - current. Masters in Climate Solutions design team
2018 - current. Governance Committee for Data Science Programs
2019. Faculty Working Group for the formation of the Division of Data Science & Information
2018 - 2019. Faculty Task Force for Data Science Minor Design
2018 - 2019. Faculty Advisory Committee of the Vice Chancellor for Undergraduate Education
2017 - 2018. ESPM Remote Sensing Faculty Search, committee member & equity liaison
2017 - 2018. ad hoc Data Science Degree Proposal Committee
2017 - 2020. Steering Committee for NSF Research Traineeship (NRT): Environment and Society: Data Science for the 21st Century (DS421)
2015 - 2019. Berkeley Research Computing Advisory Committee

National / International

2021 - current Faculty co-advisor and co-founder, Eric and Wendy Schmidt Center for Data Science and Environment, UC Berkeley.
Aug 2025 - current. National Ecological Observatory Network (NEON) Innovation Advisory Committee.
2023 - current. NSF Synthesis Center Environmental Data Science Innovation & Inclusion Lab (ESIIL) Advisory Board Member.
2019 - current. Associate Editor, *Ecology Letters*
2010 - current. rOpenSci Leadership Team
2014 - 2020. NCEAS Scientific Advisory Board
2014 - current. Rocker Project co-maintainer
2016 - 2019. Jetstream Cloud Computing Stakeholder Advisory Board, XSEDE.
Reviewer for over 40 journals, NSF review panelist, ad hoc reviewer for UC CCGA, NSF, NERSC.

Software

R Packages (on CRAN)

`roxigraph`: RDF and SPARQL for R using Oxigraph. Carl Boettiger (2026).
`rcloner`: Interface to the rclone Cloud Storage Utility. Carl Boettiger (2026).
`duckdbfs` High Performance Remote File System, Database and Geospatial Access Using duckdb. Carl Boettiger (2023).
`earthdatalogin` NASA EarthData Access Utilities. Carl Boettiger, Luis López, Yuvi Panda, Brianna Lind (2023).
`minioclient`: A High Performance Interface to S3-based Object Stores. Carl Boettiger (2023).
`gbifdb`: A High Performance Interface to GBIF. Carl Boettiger (2022).
`neonstore`: A Fast and Provenance Aware Local NEON Data Store.
Carl Boettiger, Quinn Thomas, Christine Laney, Claire Lunch (2020).
`contentid`: An Interface for Content-Based Identifiers. Carl Boettiger and Jorrit Poelen (2020).
`taxalight`: A Lightweight and Lightning-Fast Taxonomic Naming Interface. Carl Boettiger and Kari Norman (2020)
`sarsop`: Approximate POMDP Planning Software. Carl Boettiger and Jeroen Ooms and Milad Memarzadeh (2020).
`taxadb`: A High-Performance Local Taxonomic Database Interface. Carl Boettiger and Kari Norman and Jorrit Poelen and Scott Chamberlain, (2020).
`emld`: Ecological Metadata as Linked Data. Carl Boettiger (2019).
`virtuoso`: Interface to Virtuoso using ODBC. Carl Boettiger (2019).
`rdflib`: Tools to Manipulate and Query Semantic Data. Carl Boettiger (2018).
`codemetar`: Generate CodeMeta Metadata for R Packages. Carl Boettiger, Maëlle Salmon (2018).
`codemeta`: A Smaller codemetar Package. Carl Boettiger, Maëlle Salmon (2021).
`EcoNetGen`: Simulate and Sample from Ecological Interaction Networks. Marcus de Aguiar, Erica Newman, Mathias Pires, Carl Boettiger (2018).
`piggyback`: Managing Larger Data on a GitHub Repository. Carl Boettiger (2018).
`arkdb`: Archive and Unarchive Databases Using Flat Files. Carl Boettiger (2018).
`EML`: Read and Write Ecological Metadata Language File. Carl Boettiger, Matt Jones (2016; v2 2019).
`RNeXML`: Semantically Rich I/O for the NeXML Phylogenetics Format. Carl Boettiger, Scott Chamberlain, Hilmar Lapp, Rutger Vos (2014).
`pmc`: Phylogenetic Monte Carlo. Carl Boettiger (2012).
`knitcitations`: Citations for Knitr Markdown Files. Carl Boettiger (2012).
`rfishbase`: R Interface to FishBase. Carl Boettiger, Scott Chamberlain, Duncan Temple Lang, Peter Wainwright (2011; v2 2015; v3 2019).
`treebase`: Discovery, Access and Manipulation of TreeBASE Phylogenies. Carl Boettiger, Duncan Temple Lang (2011).

Python Modules (on PyPI)

`jupyter-geoagent`: A JupyterLab extension providing an agentic interface to cloud-native geospatial data. Carl Boettiger (2026).
`jupyter-sidekick`: A JupyterLab extension for working alongside language models. Carl Boettiger (2025).
`gym_climate`: A Simulation Environment for Deep Reinforcement Learning for climate policy scenarios. Carl Boettiger and Marcus Lapeyrolerie (2021).
`gym_fishing`: A Simulation Environment for Deep Reinforcement Learning for sustainable fisheries management. Carl Boettiger and Marcus Lapeyrolerie (2021).
`gym_conservation`: A Simulation Environment for Deep Reinforcement Learning for conservation scenarios. Carl Boettiger (2021)

Other Software

Carl Boettiger et al. GLEN: an open source geospatial decision-support platform using open-weight language models over cloud-native geospatial data. (Languages: Python, R)
Carl Boettiger, Dirk Eddelbuettel. The Rocker Project: Docker images for the R environment. `https://rocker-project.org`. (Language: Dockerfile)
Carl Boettiger, Matt Jones, et al. The CodeMeta Project: Software Metadata Exchange `https://codemeta.github.io` (Language: JSON-LD)

Media interviews

Martin, Glen. (2019)."How Algorithms Could Save the Planet." *California Magazine*. `https://alumni.berkeley.edu/california-magazine/just-in/2019-02-01/how-algorithms-could-save-planet`
Seltenrich, N. (2016). "Scaling the Heights of Data Science." *Breakthroughs Magazine*. `https://nature.berkeley.edu/breakthroughs/opensci-data`
Tachibana, C. (2014). "The paperless lab" *Science* 345(6195) pp. 468-470. 10.1126/science.opms.p1400087
Mascarelli, A. (2014) "Research tools: Jump off the page." *Nature* 507, 523–525. doi:10.1038/nj7493-523a
Check Hayden E (2013). "Mozilla Plan Seeks to Debug Scientific Code." *Nature*, 501, pp. 472-472. doi:10.1038/501472a
Van Noorden R (2013). "Data-Sharing: Everything on Display." *Nature*, 500, pp. 243-245. doi:10.1038/nj7461-243a
Gewin, Virginia (2013). "Turning Point: Carl Boettiger" *Nature*, 493 p 711 doi:10.1038/nj7434-711a
Wald, Chelsea (2010). "Scientists Embrace Openness" *Science*. doi:10.1126/science.caredit.a1000036

Updated August 6, 2026